

Muhammad Fahim Faisal

Autonomy & robotics engineer — perception, control, embedded

fh1m.faisal.work@gmail.com

github.com/fh1m

Dhaka, Bangladesh

Autonomy engineer working across underwater, aerial and ground robots — from the firmware on the board to the neural net that sees. Engineering lead on BRAC University's RoboSub AUV team; builder of an independent ROS 2 AUV stack with a 500 Hz control loop and Hailo-8 vision. I care about the seam between software and physics, and about numbers that ship with the measurement that produced them.

SKILLS

LANGUAGES	Python · C · C++ · Rust · JavaScript
ML	PyTorch · CUDA · TensorRT · OpenCV · YOLO11 · scikit-learn
ROBOTICS	ROS 2 · MAVLink · Nav2 · Gazebo · Pixhawk · Jetson · Hailo-8
WEB	SvelteKit · React · Next.js · Node · Flask · Django
TOOLS	Linux · Git · Docker · Bash · KiCad · PCB design

EXPERIENCE

BRACU Duburi

Oct 2023 — Sep 2026

Team Lead — Engineering · former Sub-Team Lead, AI & Machine Vision · Dhaka

- Rebuilt the vision stack on state-of-the-art CNN architecture with a custom dataset pipeline.
- Created a training pipeline that streamlines custom models with advanced hyperparameter control.
- Integrated Domain-Specific Pretraining and Source-Free Domain Adaptation.
- Deployed on edge devices for efficient on-vehicle inference.
- Built in-house dev tooling for management and deployment.

BRACU Mongol-Tori

Dec 2024 — May 2025

Contributor — Autonomous Subteam · Dhaka

- Implemented inverse kinematics for a 2-DOF arm for autonomous typing tasks.
- Built and deployed a lightweight vision unit for arm alignment, separate from the rover.
- Wrote custom OCR + image processing to identify and differentiate keys.
- Built an in-house GUI and sim environment for calibration and early testing.
- Managed the dataset pipeline for URC 2025.

Amateur Experimental Rocketry (AERD)

Feb 2023 — Feb 2025

Core Member — A&GNC, Trajectory Prediction, Parachute & Procurement · Dhaka

- Implemented a ROS 2 (Humble) stack for data acquisition.
- Worked with flight controllers (Pixhawk, Betaflight); led V1 of an in-house flight computer for a monocopter UAV.
- Worked on a VSLAM software stack for GPS-denied autonomous navigation.
- Ran trajectory prediction with RocketPy against real flight-test data.

BRACU Rocketry — LASSET Lab

Sep 2024 — Feb 2025

Sub-Team Member — Controls & Trajectory Prediction · BRAC University

- Built an in-house PID controller for a TVC gimbal; submitted the demo to the Research Excellence Awards Initiative for funding.
- Implemented a 2-axis controller.
- Built a custom GUI for tracking and monitoring.

Dubotech

Oct 2025 — Feb 2026

Jr Autonomous Engineer · Dhaka

- Developed custom firmware for sonar tech.
- Created a navigation stack for deep-water monitoring.
- Built and maintained backends for the sonar stack; started in-house tooling documentation.

Thinkit Inc

Mar 2021 — Jun 2021

Web Dev Intern · Dhaka

- Wrote back-end code for a transition to MongoDB.
- Fixed front-end bugs.

ICT Club — SOS Hermann Gmeiner College

Jan 2022 — Sep 2022

General Secretary · Dhaka

- Ran the club end to end — seminars, programs, operations.
- Automated content management with the YouTube / Facebook APIs.

EDUCATION

BRAC University

Sep 2023 — present

BSc Computer Science — Minor: Electrical & Electronic Engineering · Dhaka, Bangladesh

Currently in the 6th semester.

IU International University of Applied Sciences

2024

BSc — Autonomous Systems · Bad Honnef, Germany

One semester, on campus.

COMPETITION

2023 RoboSub 2023 — BRACU Duburi. Competed at RoboSub, San Diego — one of the few teams from Bangladesh on the world stage.

2025 RoboSub 2025 — BRACU Duburi. Direct-qualified for the semi-finals; won the special “Entrepreneurship” award.

2025 University Rover Challenge 2025 — BRACU Mongol-Tori. Ranked among the global top 10.

2026 RoboSub 2026 — BRACU Duburi. Qualified two underwater vehicles for the semi-finals.

PUBLICATIONS

Color is Also Class Evidence: Re-evaluating Domain Generalization for Underwater Object Detection

T. M. Wasif Ishmam, Yaad Kamrul Bari, Muhammad Fahim Faisal, Md Khalilur Rhaman · OpenReview, 2025

E-Commerce Recommendation System Based on Demand Clues

I. N. Reeve, T. Alam, M. Rahman Munia, A. Atif Showmik, M. F. Faisal · Universe International Journal of Interdisciplinary Research, Vol. 4 Issue 11, 2024